

Aryaman Sharma

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EDUCATION

University of Waterloo

Waterloo, ON

B.C.S Bachelor of Computer Science Honours Co-op

Expected April 2030

- Awards: President's Scholarship of Distinction | 1x Term Distinction
- Coursework: Data Structures & Algorithms, Designing Functional Programs, Tools for Software Development

EXPERIENCE

Tutor - Mathematics & Computer Science

September 2023 - June 2025

Fletcher's Meadow Secondary School

Brampton, ON

- Delivered **150+ hours** of support in computer science, teaching core **object oriented programming** concepts in **Python/Java** and guiding peers through coding projects, which boosted coding proficiency and problem solving.
- Tutored **15+ peers** in **calculus** by simplifying complex concepts like **limits, derivatives, and integrals** into accessible explanations, reinforcing students' strategies while leading to **20-30% improvements** in test scores.

PROJECTS

Heart Disease Prediction | *Python, scikit-learn, NumPy, Pandas, Git*

[GitHub](#)

- Developed an end-to-end machine learning pipeline to predict heart disease risk. achieving **85% classification accuracy**, using logistic regression on **300+ patient records**.
- Improved model reliability through **data cleaning** and **feature engineering**, validated via an **80/20 train-test split**, using Python, Pandas, NumPy, and scikit-learn.
- Evaluated model performance on a held-out test using accuracy and **classification metrics**, ensuring robust generalization, with scikit-learn's evaluation framework.

Car Marketplace | *Java, Git*

[GitHub](#)

- Built a **Java based car marketplace** using **object-oriented design** to support **50+ vehicle listings**, real-time search/filtering, and end-to-end buy/sell transactions.
- Improved maintainability by designing a **modular architecture** with **15+ reusable classes** and interfaces, cutting feature development time by **30%** and reducing code duplication.

Portfolio Website | *React, JavaScript, HTML/CSS, GitHub Pages*

[GitHub](#)

- Designed and developed a fully responsive personal portfolio using **React and JavaScript**, showcasing projects, my information, and contact details in an organized UI.
- Improved performance and accessibility through reusable components, semantic HTML, optimized assets, and mobile-first CSS layouts, ensuring smooth **cross-device rendering**.
- **Deployed** and maintained the site via GitHub Pages with **Git-based version control**, enabling seamless updates and continuous improvement.

RC Vehicle | *C++, Arduino, Embedded Systems, Electronics*

[GitHub](#)

- Designed and built an RC car using Arduino microprocessor, C++ firmware using structured, modular code and interrupt-based logic, achieving a top speed of **15 km/h** with low-latency real-time motor control.
- Integrated **embedded systems** components for wireless operation, delivering a **30m control range** with stable signal performance and reliable system responsiveness.
- Developed firmware using **iterative Agile-style prototyping** and **testing** across multiple development cycles to improve **vehicle performance by 25%**.

Ecommerce Store + Dashboard | *TypeScript, ReactJS, NextJS, REST APIs, Prisma, Stripe*

[GitHub](#)

- Developed an **e-commerce dashboard** enabling users to create and manage multiple stores, each equipped with **custom REST APIs for store management**, order tracking, and product addition, including administrative controls and advanced analytics capabilities.
- Created an e-commerce store featuring product **information sourced dynamically from the dashboard**, equipped with **filtering options** based on size, category, and more.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript, SQL (MySQL), HTML, CSS, Racket

Technologies: Git, Pandas, Scikit-learn, NumPy, Pygame, Matplotlib, React, NodeJS, AWS Lambda, Spring Boot

Developer Tools: GitHub, VS Code, IntelliJ IDEA, PyCharm, Eclipse, Linux, Vim, Power BI, Microsoft 365